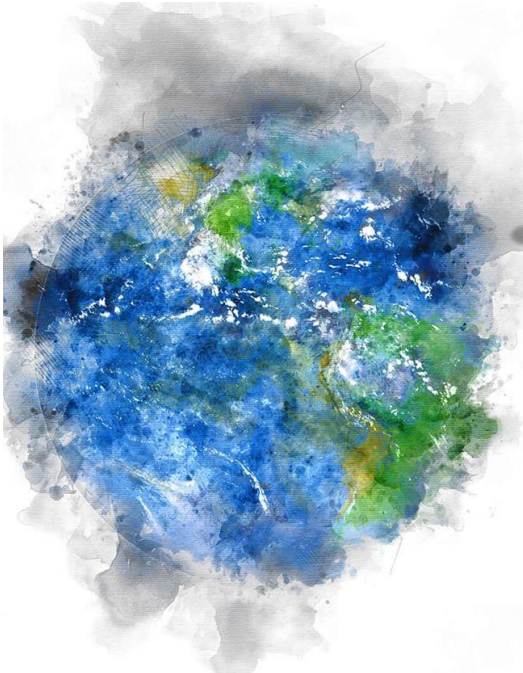




CLASSIFICATION IN DATA MINING

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OUTLINE :

1. *Basic concepts of data mining classification*
2. *Classification Technique*

- ✓ *Decision Tree*
- ✓ *Artificial Neural Network (see you in the ANN course) ☺☺ ☺☺*
- ✓ *Nearest-Neighbor Classifier*
- ✓ *Naïve Bayes Classifier*
- ✓ *Support Vector Machine (SVM)*

Basic Concepts Of Data Mining Classification

- Bagian sangat penting dalam data mining adalah teknik klasifikasi, yaitu bagaimana mempelajari sekumpulan data sehingga dihasilkan aturan yang bisa mengklasifikasi atau mengenali data-data baru yang belum pernah dipelajari.
- Klasifikasi merupakan pekerjaan menilai objek data untuk memasukkannya ke dalam kelas tertentu dari sejumlah kelas yang tersedia.
- Terdapat dua pekerjaan utama dalam klasifikasi :
 - 1) Pembangunan model sebagai prototipe untuk disimpan sebagai memori.
 - 2) Penggunaan model tersebut untuk melakukan pengenalan/klasifikasi/ prediksi pada suatu objek data lain sehingga dapat diketahui di kelas mana objek data tersebut dalam model yang sudah disimpannya.

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CLASSIFICATION : DEFINITION

- » Klasifikasi dapat didefinisikan sebagai proses u/ menyatakan suatu objek data sebagai salah satu kategori (kelas) yang telah didefinisikan sebelumnya (Zaki et al. 2013).
- » Klasifikasi didefinisikan sebagai pekerjaan yang melakukan pelatihan /pembelajaran terhadap fungsi target f yang memetakan setiap set atribut (fitur) x ke satu dari sejumlah label kelas y yang tersedia.
- » Classification is a data mining function that assigns items in a collection to target categories or classes.
- » The goal of classification is to accurately predict the target class for each case in the data.
- » For example, a classification model could be used to identify loan applicants as *low, medium, or high credit risks*.

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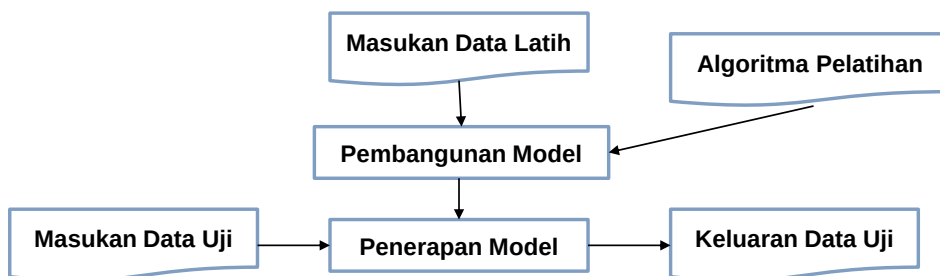
CLASSIFICATION : DEFINITION

- » Given a collection of records (*training set*)
Each record contains a set of *attributes*, one of attributes is the *class*.
- » Find a *model* for class attribute as a function of the values of other attributes.
- » Goal : previously unseen records should be assigned a class as accurately as possible.
A *test set* is used to determine the accuracy of the model. Usually, the given data set used to build the model and test set used to validate it.

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CLASSIFICATION : DEFINITION

- » Model dalam klasifikasi seperti kotak hitam, yang dapat menerima masukan, kemudian melakukan proses berpikir atas masukan tsb, dan memberi jawaban sebagai output dari hasil proses berpikir.
- » Gambaran framework klasifikasi :



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Classification vs. Prediction

- **Classification:**
 - ✓ Predicts categorical class labels
 - ✓ Classifies data (constructs a model) based on the training set and the values (**class labels**) in a classifying attribute and uses it in classifying new data.
- **Prediction:**
Models continuous-valued functions, i.e., predicts unknown or missing values.
- **Typical Applications**
 - ✓ credit approval
 - ✓ target marketing
 - ✓ medical diagnosis
 - ✓ treatment effectiveness analysis

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Pengukuran kinerja Klasifikasi (Tingkat akurasi)

- ✓ Sebuah sistem yang melakukan klasifikasi diharapkan dapat melakukan klasifikasi semua set data dengan benar, tetapi tidak dapat dipungkiri bahwa kinerja suatu sistem kemungkinan tidak bisa 100% benar, sehingga sistem klasifikasi perlu untuk diukur kinerjanya.
- ✓ **Confusion matrix** digunakan untuk menghitung tingkat akurasi. **Confusion matrix** terdiri dari klasifikasi actual dan predicted yang dilakukan oleh sistem klasifikasi (Tan, 2006).

	PREDICTED CLASS	
	Class = Yes	Class = No
	Class = Yes	Class = No
ACTUAL CLASS	a	b
	c	d

a: **TP** (true positive)
b: **FN** (false negative)

c: **FP** (false positive)
d: **TN** (true negative)

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Pengukuran kinerja Klasifikasi (Tingkat akurasi)

✓ Tingkat akurasi kebenaran ditunjukkan pada persamaan dibawah ini

$$\text{Akurasi} = \frac{a + d}{a + b + c + d}$$

Dimana

- **a** adalah *True Positive* yang berarti hasil kategorisasi *testing* adalah Z_i sesuai dengan data *testing* kategori Z_i .
- **b** adalah *False Positive* artinya hasil *testing* terkategori Z_i namun data kategori bukan Z_i .
- **c** adalah *False Negative* berarti hasil kategorisasi *testing* bukan Z_i , tidak sesuai data *testing* kategori Z_i .
- **d** adalah *True Negative* berarti hasil kategorisasi *testing* bukan Z_i , dan sesuai dengan data *testing* kategori yaitu bukan Z_i

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• What can we learn from this matrix?

	n=165	
	Predicted: NO	Predicted: YES
Actual: NO	50	10
Actual: YES	5	100

- There are two possible predicted classes: "yes" and "no". If we were predicting the presence of a disease, for example, "yes" would mean they have the disease, and "no" would mean they don't have the disease.
- The classifier made a total of 165 predictions (e.g., 165 patients were being tested for the presence of that disease).
- Out of those 165 cases, the classifier predicted "yes" 110 times, and "no" 55 times.
- In reality, 105 patients in the sample have the disease, and 60 patients do not.

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- Let's now define the most basic terms, which are whole numbers (not rates):

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

- true positives (TP): These are cases in which we predicted yes (they have the disease), and they do have the disease.
- true negatives (TN): We predicted no, and they don't have the disease.
- false positives (FP): We predicted yes, but they don't actually have the disease. (Also known as a "Type I error.")
- false negatives (FN): We predicted no, but they actually do have the disease. (Also known as a "Type II error.")

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- This is a list of rates that are often computed from a confusion matrix:
- Accuracy: Overall, how often is the classifier correct?
- Misclassification Rate: Overall, how often is it wrong?

$$(TP+TN)/total = (100+50)/165 = 0.91$$

$$(FP+FN)/total = (10+5)/165 = 0.09$$

equivalent to 1 minus Accuracy
also known as "Error Rate"

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

- True Positive Rate: When it's actually yes, how often does it predict yes?
 $TP/\text{actual yes} = 100/105 = 0.95$
also known as "Sensitivity" or "Recall"
- False Positive Rate: When it's actually no, how often does it predict yes?
 $FP/\text{actual no} = 10/60 = 0.17$

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- This is a list of rates that are often computed from a confusion matrix:
- Specificity: When it's actually no, how often does it predict no?
 $TN/\text{actual no} = 50/60 = 0.83$
 equivalent to 1 minus False Positive Rate
- Precision: When it predicts yes, how often is it correct?
 $TP/\text{predicted yes} = 100/110 = 0.91$
- Prevalence: How often does the yes condition actually occur in our sample?
 $\text{actual yes}/\text{total} = 105/165 = 0.64$

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A Sample Classification Problem

- Suppose you want to predict which of your customers are likely to increase spending if given an affinity card. You could build a model using demographic data about customers who have used an affinity card in the past. Since we want to predict either a positive or a negative response (will or will not increase spending), we will build a binary classification model.
- This example uses classification model, `dt_sh_clas_sample`, which is created by one of the Oracle Data Mining sample programs (described in [Oracle Data Mining Administrator's Guide](#)).
- Figure 1 shows six columns and ten rows from the case table used to build the model. A target value of 1 has been assigned to customers who increased spending with an affinity card; a value of 0 has been assigned to customers who did not increase spending.

✓ Figure 1. **Sample Build Data for Classification**

case ID	predictors				target
CUST_ID	CUST_GENDER	EDUCATION	OCCUPATION	AGE	AFFINITY_CARD
101501	F	Masters	Prof.	41	0
101502	M	Bach.	Sales	27	0
101503	F	HS-grad	Cleric.	20	0
101504	M	Bach.	Exec.	45	1
101505	M	Masters	Sales	34	1
101506	M	HS-grad	Other	38	0
101507	M	< Bach.	Sales	28	0
101508	M	HS-grad	Sales	19	0
101509	M	Bach.	Other	52	0
101510	M	Bach.	Sales	27	1

After undergoing testing (see "Testing a Classification Model"), the model can be applied to the data set that you wish to mine.

- ✓ Figure 2 shows some of the predictions generated when the model is applied to the customer data set provided with the Oracle Data Mining sample programs. It displays several of the predictors along with the prediction (1=will increase spending; 0=will not increase spending) and the probability of the prediction for each customer.

Classification Results

Activity: KLT_MINING_DATA_BU380101404_AA: Result ...

File Publish Help

Apply Output Apply Settings Task

Apply Output Table: MINING_DATA_APPL189328013_A

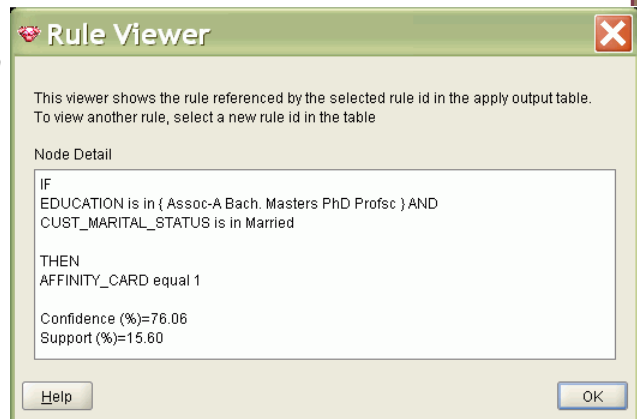
Fetch Size: 100 Refresh

DMR\$CASE_ID	OCCUPATION1	CUST_GENDER1	CUST_MARITAL...	PREDICTION	PROBABILITY	NODE
100,001	Exec.	F	Widowed	0	0.9192	10
100,002	Prof.	F	NeverM	0	0.7619	8
100,003	Sales	M	NeverM	0	0.9924	9
100,004	Sales	F	Divorc.	0	0.9192	10
100,005	Crafts	M	Married	1	0.7606	5
100,006	Prof.	F	NeverM	0	0.9924	9
100,007	Other	F	Divorc.	0	0.9924	9
100,008	Crafts	M	NeverM	0	0.9924	9
100,009	Prof.	M	Married	1	0.7606	5
100,010	Crafts	M	Married	0	0.6838	6

Rule...

- ✓ Since this classification model uses the Decision Tree algorithm, rules are generated with the predictions and probabilities.
- ✓ Figure 3 shows the rule for node 5. The rule states that married customers who have a college degree (Associates, Bachelor, Masters, Ph.D., or professional) are likely to increase spending with an affinity card.

✓ **Decision Tree Rules for Classification**



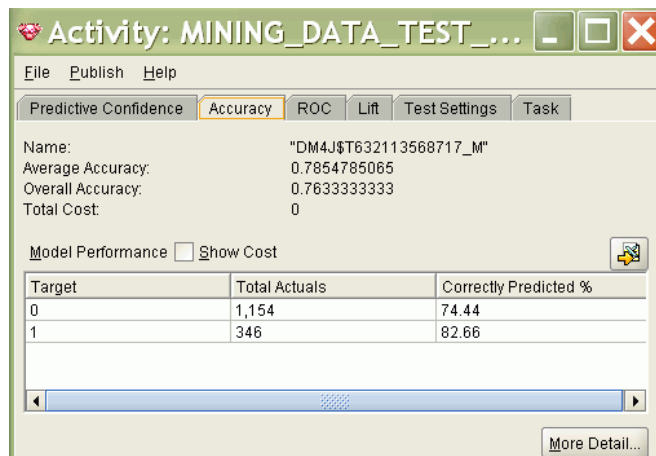
Testing a Classification Model

- ✓ A classification model is tested by applying it to test data with known target values and comparing the predicted values with the known values.
- ✓ The test data must be compatible with the data used to build the model and must be prepared in the same way that the build data was prepared. Typically the build data and test data come from the same historical data set. A percentage of the records is used to build the model; the remaining records are used to test the model.
- ✓ Test metrics are used to assess how accurately the model predicts the known values. If the model performs well and meets the business requirements, it can then be applied to new data to predict the future.

Accuracy

- ✓ Accuracy refers to the percentage of correct predictions made by the model when compared with the actual classifications in the test data.
- ✓ Figure 4 shows the accuracy of a binary classification model in Oracle D Miner.

- ✓ Figure 4 the accuracy of a binary classification model



Confusion Matrix

- A confusion matrix displays the number of correct and incorrect predictions made by the model compared with the actual classifications in the test data. The matrix is n -by- n , where n is the number of classes.
- Figure 5 shows a confusion matrix for a binary classification model. The rows present the number of actual classifications in the test data. The columns present the number of predicted classifications made by the model.

		PREDICTED CLASS	
		affinity_card = 1	affinity_card = 0
ACTUAL CLASS	affinity_card = 1	516	25
	affinity_card = 0	10	725

- In this example, the model correctly predicted the positive class for affinity_card 516 times and incorrectly predicted it 25 times. The model correctly predicted the negative class for affinity_card 725 times and incorrectly predicted it 10 times.
- The following can be computed from confusion matrix:
 - ✓ The model made 1241 **correct predictions** ($516 + 725$).
 - ✓ The model made 35 **incorrect predictions** ($25 + 10$).
 - ✓ There are 1276 **total scored cases** ($516 + 25 + 10 + 725$).
 - ✓ The **error rate** is $35/1276 = 0.0274$.
 - ✓ The **overall accuracy rate** is $1241/1276 = 0.9725$.

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SEE U ...
CLASSIFICATION METHODS

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